Vishal S

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Summary

Computer Science graduate and software engineer experienced in production Ruby on Rails, Python APIs (Django, FastAPI), and Docker/Kubernetes CI/CD and GitOps pipelines, with applied machine learning for geospatial analysis.

Education

Coimbatore Institute of Technology May 2026
Bachelor of Engineering in Computer Science **CGPA: 8.86**

Skills

Programming Languages: Ruby, Python, JavaScript, TypeScript, C

Frontend: HTML, CSS, Tailwind CSS, React, Streamlit

Web Frameworks: Ruby on Rails, Django, Flask, FastAPI, React, AngularJS

Databases: PostgreSQL, MySQL, SQLite3, Firebase Firestore

AI & ML: Claude API, LangChain, Pinecone, Hugging Face, Scikit-learn, Autoencoders, GeoPandas, Plotly

DevOps & Cloud: Docker, Kubernetes, Argo CD, GitHub Actions, Kustomize, Trivy, DigitalOcean, OpenVPN

Tools: Git, GitHub, GitLab, Azure DevOps, Jira

Experience

Rootquotient Technologies **Dec 2025 – Present**
SDE (Jun 2026 – Present) | SDE Intern (Dec 2025 – May 2026) *Chennai, India*

- Engineered and delivered production features on a legacy Ruby on Rails enterprise pension application, including cron job scheduling, MFA integration and Ping Authentication workflows serving thousands of plan members.
- Redesigned the pension application flow with targeted UX improvements, reducing partial/incomplete application submissions by over 75%.
- Eliminated a critical database inconsistency where pre-approval registrations persisted alongside approved user records, achieving 100% data integrity for the affected flow.
- Supported Docker-based containerization of the on-premise environment, coordinating across database, infrastructure, and DevOps dependencies via VPN and remote desktop to ensure zero-downtime delivery.

Sálim Ali Centre for Ornithology and Natural History (SACON) **Jan 2025 – Feb 2025**
Software Engineering Intern *Coimbatore, India*

- Developed a machine learning approach to habitat suitability analysis for hornbills using autoencoders and geospatial visualization.
- Processed satellite and field data to create interactive maps identifying priority conservation areas.
- Integrated computational methods with ecological research to support biodiversity conservation.

Projects

Pension Plan Member Portal | *Ruby on Rails, Docker, MFA* **Dec 2025 – Present**

- Contributed to a legacy enterprise application handling pension and online registration flows for a public-sector pension plan.
- Integrated MFA and Ping Authentication, managed scheduled jobs, resolved production edge cases, and supported migration to a containerized Docker environment.

Hornbill Habitat Suitability Analysis | *Python, Scikit-learn, Autoencoders, GeoPandas | GitHub* **Mar 2025**

- Built a machine learning pipeline to model habitat suitability of hornbill species using ecological, telemetry, and climate data.
- Identified core habitat zones and movement behavior, and visualized spatial patterns on interactive maps with satellite imagery layers.

Socio-API | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL | GitHub* **Sept 2024**

- Developed a social media API supporting multi-account management, posts, and user profiles with role-based access control.
- Implemented secure JWT authentication and authorization to safeguard user data.
- Used SQLAlchemy and Alembic for database interactions and scalable schema migrations.

Additional Projects

Tetris GitOps Pipeline | *Docker, GitHub Actions, Trivy, Kubernetes, Argo CD, Kustomize | GitHub* **Aug 2026**

- Built a CI/CD and GitOps pipeline around a forked open-source JavaScript game: GitHub Actions auto-versions releases, scans with Trivy, and publishes the image to Docker Hub.
- Deployed to a Kubernetes cluster with Argo CD and Kustomize using a pull-based model, so no cluster credentials are stored in GitHub; rollbacks are done through Git.
- Hardened the container (non-root user, read-only filesystem, dropped capabilities) and documented the architecture and runbook.

Fun-Gram | *Vite, React, Chakra UI, Firebase | GitHub* **Jan 2025**

- Built an Instagram-style social media app with real-time posting, liking, and commenting.
- Integrated Firebase Authentication, Firestore, and Storage; designed a responsive UI with Chakra UI and deployed on Firebase Hosting.

Easy-Kart | *HTML, CSS, JavaScript, Django, Stripe | GitHub* **Jul 2024**

- Built a full-stack e-commerce website with product listings, shopping cart, and secure checkout using Stripe.
- Enabled users to register, manage profiles, list products for sale, and make purchases.

OpenVPN on DigitalOcean Guide | *OpenVPN, DigitalOcean | GitHub* **Jul 2024**

- Authored a step-by-step guide with screenshots for connecting to a DigitalOcean-hosted OpenVPN server using OpenVPN Connect, including verifying the new IP address after connecting.